Lecture 22:
normalizing flows

November 9, 2026

final project: Apply a method from this course to an astro/physics problem of your choosing.

It can be your own research data, public survey data, or a simulator you write. It must be new work done for this course.

suggestions: show validation, compare to baseline, make it reproducible (GitHub)

*one paragraph proposal (**Nov 16**)*

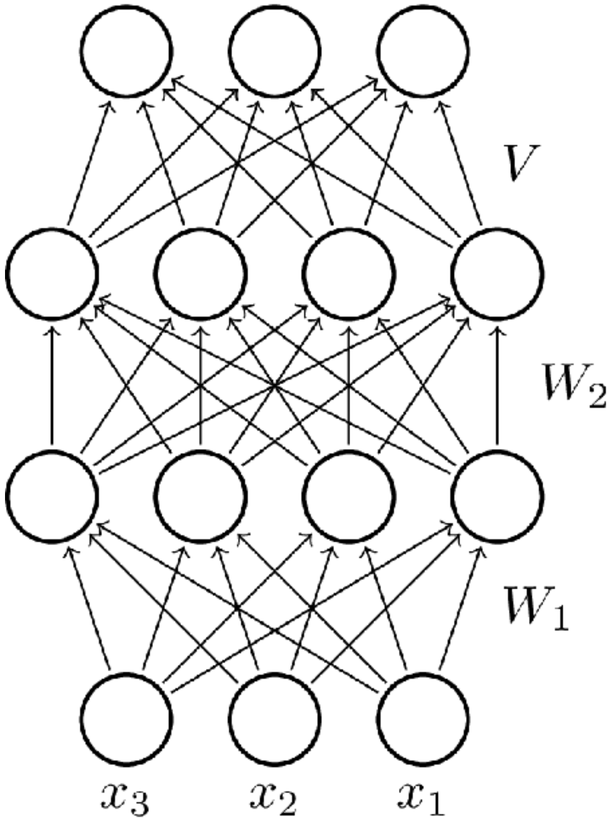
*10 minute talk (**Dec 2/7**)*

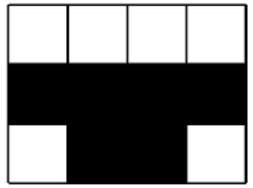
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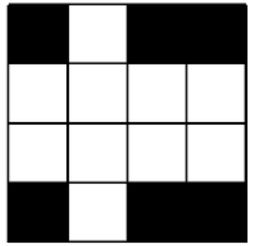
*proposal (**Nov 16**): one paragraph covering*

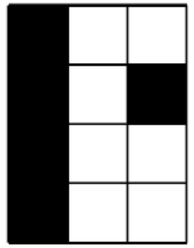
- *scientific question*
- *data/simulation*
- *method*
- *validation*

recap: MADE

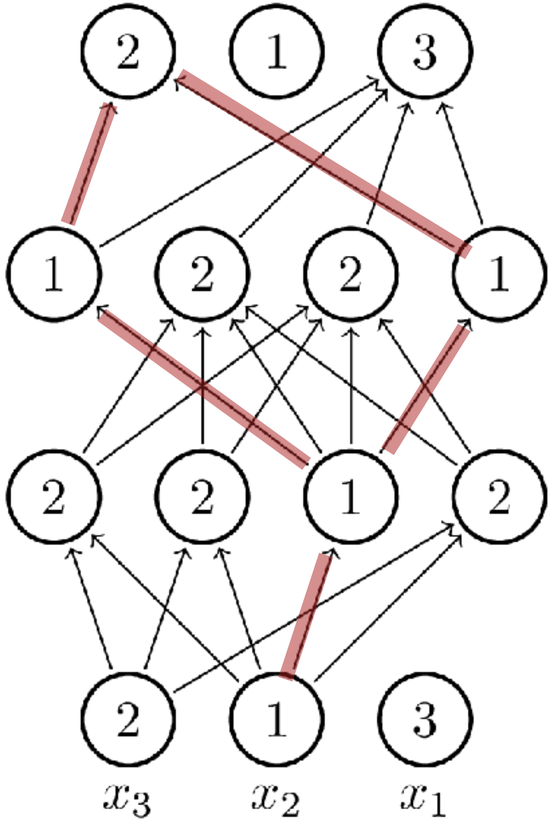


$M_V =$ 

$M_{W_2} =$ 

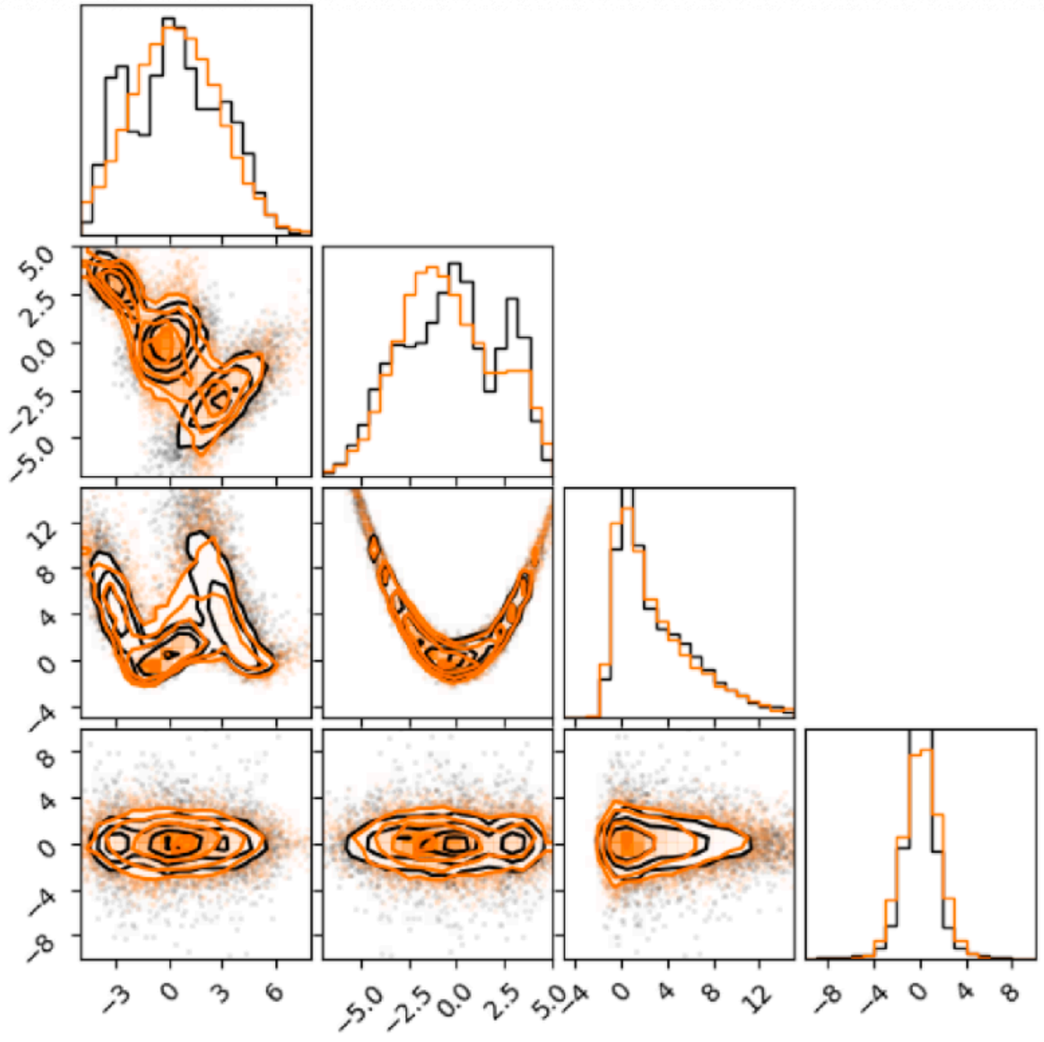
$M_{W_1} =$ 

$p(x_3 \mid x_2)$ $p(x_2)$ $p(x_1 \mid x_2, x_3)$



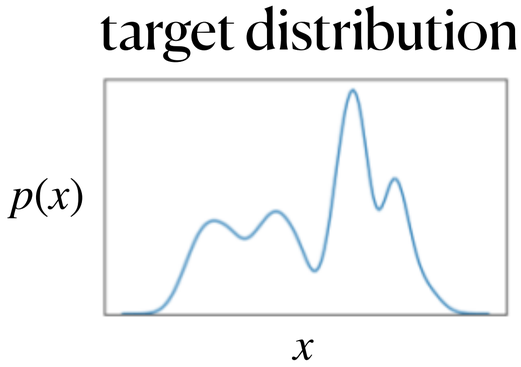
$\times$ **masks**

recap: MADE



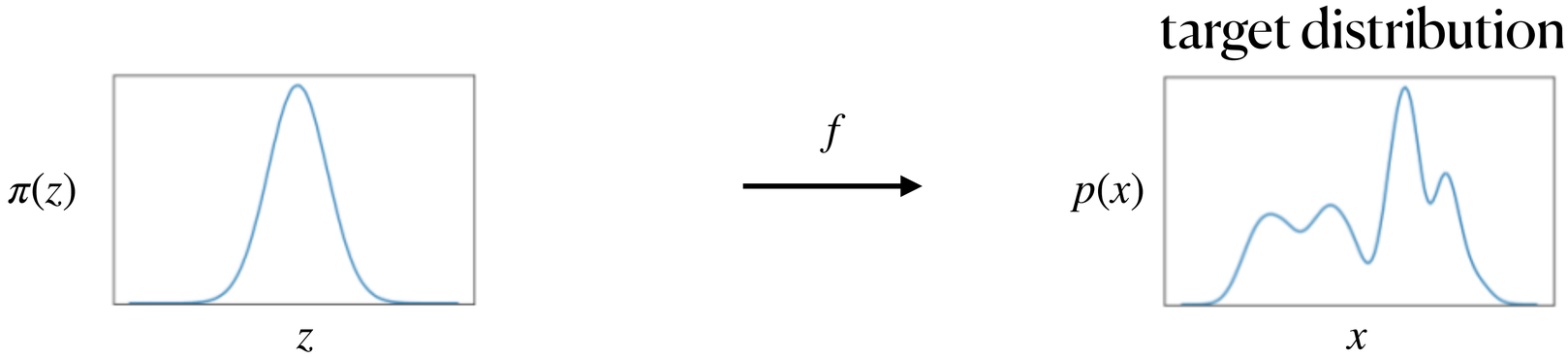
neural density estimation: **normalizing flows**

normalizing flows



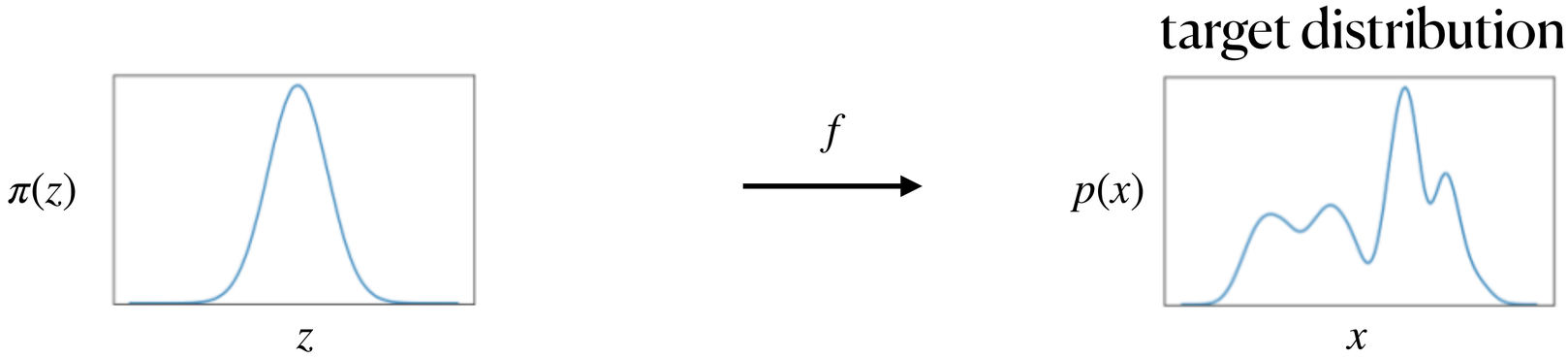
https://gebob19.github.io/normalizing-flows/

normalizing flows: invertible transformation $f : z \rightarrow x$ between our target $p(x)$ and a simple base distribution?



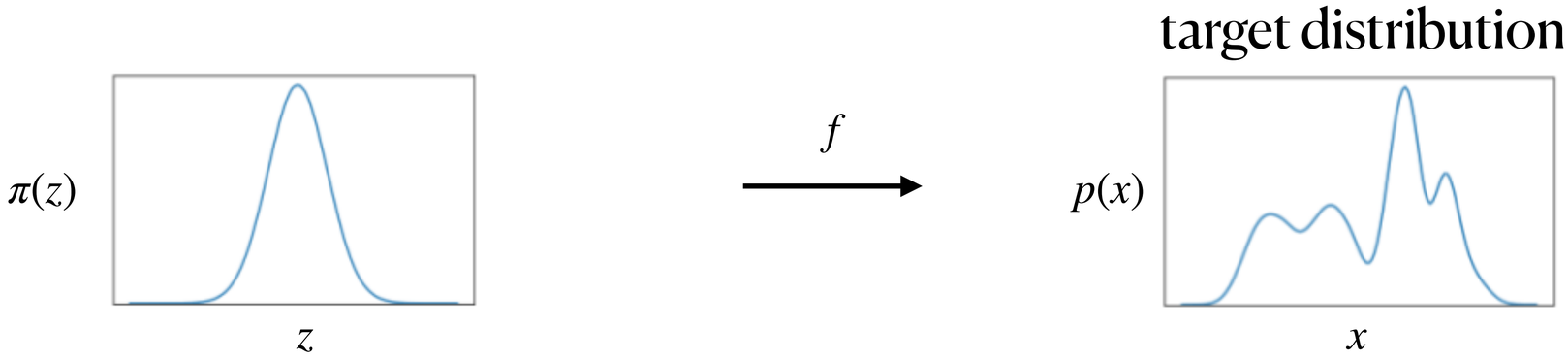
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evaluate with change of variables: $p(x) = \pi(f^{-1}(x)) \left| \det \left(\frac{\partial f^{-1}}{\partial x} \right) \right|$

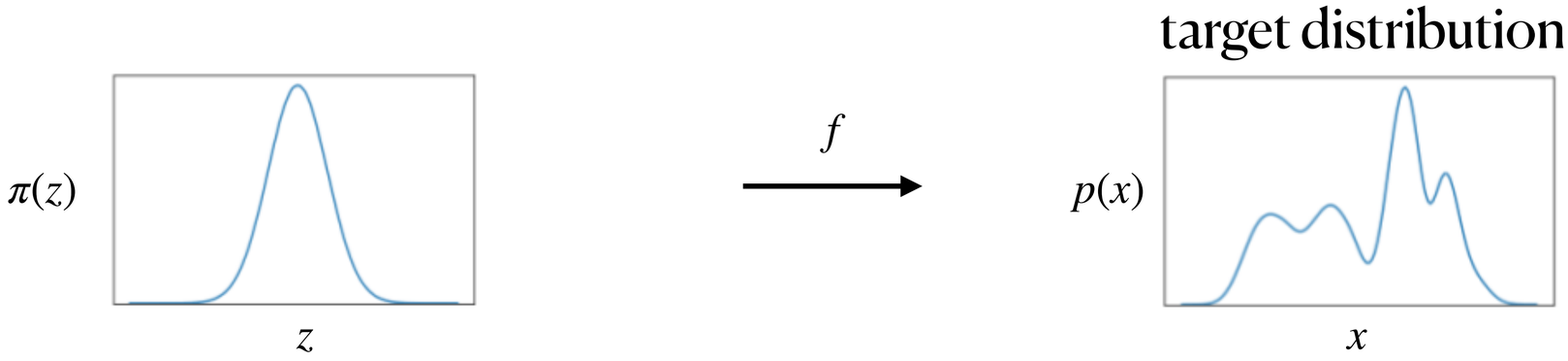
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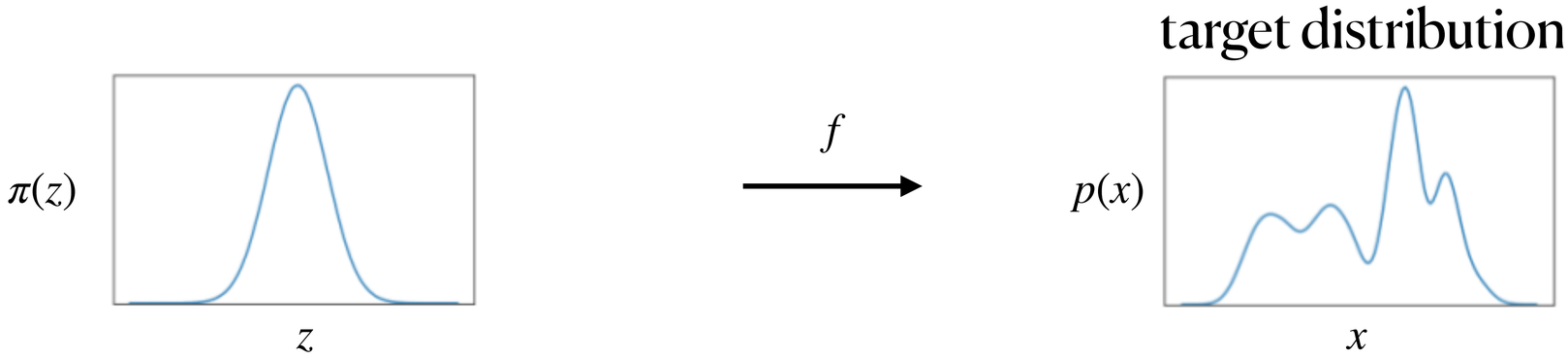


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f is easy to invert and the determinant of the Jacobian is easy to compute

normalizing flows can be **stacked** for more flexibility — *i.e.* approximate more complex distributions



$$f = f_1 \cdot f_2 \cdot \ldots \cdot f_n$$

$f_1, f_2 \ldots f_n$ are invertible, then f is invertible

e.g. planar flows

$$\mathbf{x} = \mathbf{z} + \mathbf{u}\ g(\mathbf{w}^T\mathbf{z} + b)$$

$\mathbf{u}$, $\mathbf{w}$, b are free parameters

$g = \tanh$

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$$\frac{\partial f(\mathbf{z})}{\partial \mathbf{z}} = \mathbf{I} + \mathbf{u}\ g'(\mathbf{w}^T\mathbf{z} + b)\,\mathbf{w}^T$$

$$\det \frac{\partial f(\mathbf{z})}{\partial \mathbf{z}} = 1 + g'(\mathbf{w}^T\mathbf{z} + b)\,\mathbf{w}^T\mathbf{u}$$

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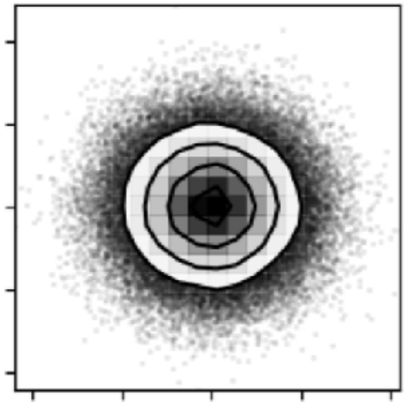
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```
w = np.array([5., 0])
u = np.array([1., 0])
b = 0
```



$\pi(\mathbf{z})$

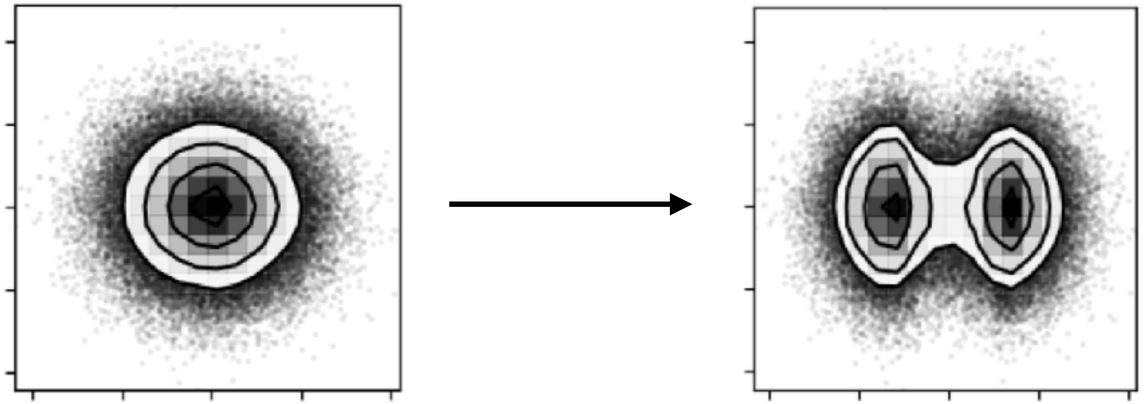
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$\pi(\mathbf{z})$ $p(\mathbf{x})$

masked autoregressive flows (MAF) —

Papamakarios et al. (2017)

MAF — autoregressive models as flows

$$p(\mathbf{x}) = \prod_{i=1}^{N} p(x_i \mid \mathbf{x}_{i-1:1})$$

$$\begin{aligned} p(\mathbf{x}) = \; & p(x_N \mid x_1, x_2, \ldots x_{N-2}, x_{N-1}) \times \\ & p(x_{N-1} \mid x_1, x_2, \ldots x_{N-2}) \times \\ & \cdots \\ & p(x_3 \mid x_1, x_2) \times \\ & p(x_2 \mid x_1) \times \\ & p(x_1) \end{aligned}$$

Chain rule

MAF — autoregressive models as flows

$$p(\mathbf{x}) = \prod_{i=1}^{N} p(x_i \mid \mathbf{x}_{i-1:1})$$

$$p(x_i \mid \mathbf{x}_{i-1:1}) = \mathcal{N}(x_i \mid \mu_i, (\exp \alpha_i)^2) \qquad \text{where} \quad \begin{aligned} \mu_i &= \mathrm{NN}_{\mu_i}(\mathbf{x}_{i-1:1}) \\ \alpha_i &= \mathrm{NN}_{\alpha_i}(\mathbf{x}_{i-1:1}) \end{aligned}$$

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we can sample $x_i \sim p(x_i \mid \mathbf{x}_{i-1:1})$ as

$$x_i = z_i \exp \alpha_i + \mu_i \text{ where } z_i \sim \mathcal{N}(0,1)$$

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also easy to invert: $u_i = (x_i - \mu_i) \exp(-\alpha_i)$ and evaluate $\left| \det \frac{\partial f^{-1}}{\partial \mathbf{x}} \right| = \exp\left(- \sum_i \alpha_i \right)$

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MAF = stacking MADEs

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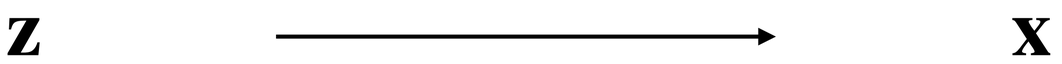
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MAF = stacking MADEs — change the order between them

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$$x_i = \mu_i(x_{i-1:1}) + \exp(\alpha_{i-1:1})\ z_i$$



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$$y_i = \mu_i(x_{i-1:1}) + \exp(\alpha_{i-1:1})\ z_i$$

2 MADE in **same order**

$$x_i = \mu'_i(x_{i-1:1}) + \exp(\alpha'_{i-1:1})\ y_i$$

z ⟶ **y** ⟶ **x**

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bigger network for μ, α but not a more flexible NDE

z ⟶ **y** ⟶ **x**

MAF = stacking MADEs — change the order between them



in-class: SBI with neural density estimation and MADE

ast384c/inclass/lec21_nde/nde_made.ipynb